Т5 $\xi \sim R(\theta, 2\theta)$ $\qquad p(x,\theta) = \begin{cases} \frac{1}{\theta}, & x \in [\theta, 2\theta] \\ 0, & x \notin [\theta, 2\theta] \end{cases}$

$M\xi = \frac{\theta + 2\theta}{2} = \frac{3}{2}\theta$ $\qquad \overline{X_n}$ – выборка

$M\xi^2 = \int_\theta^{2\theta} \frac{1}{\theta} x^2 dx = \frac{x^3}{3\theta}\Big|_\theta^{2\theta} = \frac{7}{3}\theta^2$

$D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{28-27}{12}\theta^2 = \frac{\theta^2}{12}$

а), б)

ОММ

$\alpha_1 = \int_0^{2\theta} \frac{1}{\theta} \cdot x\, dx = M\xi = \frac{3}{2}\theta$ $\qquad \overline{X} = \frac{3}{2}\theta$

$\alpha_1 = \tilde{\alpha}_1 = \overline{X} \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\overline{X}$

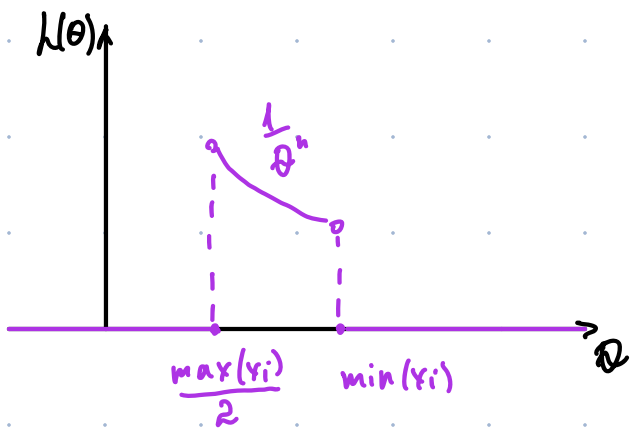
$M\tilde{\theta}_1 = \frac{2}{3} M\overline{X} = \frac{2}{3} M\xi = \frac{2}{3} \cdot \frac{3}{2}\theta = \theta \Rightarrow$ несмещенная

$D\tilde{\theta}_1 = \frac{4}{9} D\overline{X} = \frac{4}{9n} D\xi = \frac{4}{9n} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{27n} \xrightarrow[n \to \infty]{} 0 \Rightarrow$ состоят.

ОМП

$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{1}{\theta^n}, \quad x_i \in [\theta, 2\theta]$

$\theta \le x_i \le 2\theta \quad \forall i \Rightarrow \frac{\max(x_i)}{2} \le \theta \le \min(x_i)$



$\tilde{\theta}_2 = \frac{x_{max}}{2}$

$$y = \frac{x_{max} - \theta}{\theta} \Rightarrow y \sim R(0,1) \qquad x_{max} = \theta(1+y)$$

$$M x_{max} = M\left[\theta(1+y)\right] = \theta M[1+y] = \theta\left(1 + \frac{n}{n+1}\right) =$$

$$= \theta + \frac{\theta n}{n+1} = \frac{(n+1)\theta + \theta n}{n+1} = \theta \frac{2n+1}{n+1} \Rightarrow$$

$$M\tilde{\theta}_2 = \frac{1}{2} \cdot \theta \frac{2n+1}{n+1} = \frac{\theta}{2} \frac{2n+1}{n+1} \Rightarrow$$ смещенная

$$\tilde{\theta}_2' = \frac{2n+2}{2n+1} \frac{x_{max}}{2} = \frac{n+1}{2n+1} x_{max}$$

$$M\tilde{\theta}_2^2 = \frac{1}{4} \int_0^{2\theta} \frac{n}{\theta} \cdot y^2 \left(\frac{y}{\theta} - 1\right)^{n-1} dy = \left\{\frac{y}{\theta} = t\right\} = \frac{\theta^2}{4} \int_1^2 n t^2 (t-1)^{n-1} dt =$$

$$= \left\{u = t-1\right\} = \frac{\theta^2}{4} \cdot n \int_0^1 (u+1)^2 u^{n-1} du =$$

$$= \frac{\theta^2}{4} \cdot n \int_0^1 \left(u^{n+1} + 2u^n + u^{n-1}\right) du = \frac{\theta^2}{4} \cdot n \left(\frac{1}{n+2} + \frac{2}{n+1} + \frac{1}{n}\right) =$$

$$= \frac{\theta^2}{4} \cdot \frac{2(2n^2+4n+1)}{(n+1)(n+2)} = \frac{\theta^2}{2} \frac{2n^2+4n+1}{(n+1)(n+2)}$$

$$D\tilde{\theta}_2 = \frac{\theta^2}{2} \frac{2n^2+4n+1}{(n+1)(n+2)} - \frac{\theta^2(4n^2+4n+1)}{4(n+1)^2} = \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow[n \to \infty]{} 0$$

$\Rightarrow \tilde{\theta}_2$ – состоятельная

$$D(\tilde{\theta}_2') = \frac{\cancel{(2n+2)^2}}{(2n+1)^2} \frac{n\theta^2}{\cancel{4(n+1)^2}(n+2)} = \frac{n\theta^2}{(n+2)(2n+1)^2} \xrightarrow[n\to\infty]{} 0 \Rightarrow$$

$\Rightarrow$ $\tilde{\theta}_2'$ — состоятельная

c) $D\tilde{\theta}_1 = \frac{\theta^2}{27n}$ $\qquad$ $D\tilde{\theta}_2' = \frac{n\theta^2}{(n+2)(2n+1)^2}$

$$\frac{\theta^2}{27n} \geq \frac{n\theta^2}{(n+2)(2n+1)^2} \quad \text{при} \quad n \geq 3$$

$\tilde{\theta}_2'$ эффективнее $\tilde{\theta}_1$

d) Точный доверительный интервал $\quad$ β

$\vec{X}_n$ — выборка $\qquad$ $\xi \sim R[\theta, 2\theta]$

$$f(\vec{X}_n, \theta) = \frac{X_{max}}{\theta} - 1$$

$$P(f < t) = P(X_{max} < \theta t + \theta) = (F(\theta t + \theta))^n =$$

$$= \left\{ F(x) = \begin{cases} 0, & x \leq \theta \\ \frac{x}{\theta} - 1, & \theta < x \leq 2\theta \\ 1, & x > 2\theta \end{cases} \right\} = \begin{cases} 0, & t \leq 0 \\ t^n, & 0 < t \leq 1 \\ 1, & t > 1 \end{cases}$$

$\alpha + \beta = 1$

$$t_1 = q_{\frac{\alpha}{2}} = \sqrt[n]{\frac{\alpha}{2}} = t_2 = q_{1-\frac{\alpha}{2}} = \sqrt{1 - \frac{\alpha}{2}} = \sqrt[n]{\frac{1+\beta}{2}} = \sqrt[n]{\frac{1-\beta}{2}}$$

$$P\left(t_1 < \frac{X_{max}}{\theta} - 1 < t_2\right) = \beta$$

$$t_1 + 1 < \frac{X_{max}}{\theta} < t_2 + 1$$

$$\frac{1}{1+t_2} < \frac{\theta}{X_{max}} < \frac{1}{1+t_1}$$

Получаем
$$P\left(\frac{X_{max}}{1+\sqrt[n]{\frac{1+\beta}{2}}} < \theta < \frac{X_{max}}{1+\sqrt[n]{\frac{1-\beta}{2}}}\right) = \beta$$

e) Асимптотический доверительный интервал

ОММ $\quad \tilde{\theta}_1 = \frac{2}{3}\bar{X} = \frac{2}{3}\tilde{\alpha}_1 = g(\tilde{\alpha}_1) \qquad \tilde{\alpha}_1 = \bar{X} \qquad \alpha_1 = \frac{3\theta}{2}$

$$g(\alpha_1) = \frac{2}{3}\alpha_1 = \theta \qquad \nabla g = \frac{2}{3} \qquad \hat{\theta}_1 = \frac{2}{3}\bar{X}$$

$$K_{11} = D\xi = \frac{\theta^2}{12}$$

$$\sigma^2(\theta) = \left(\frac{2}{3}\right)^2 \cdot \frac{\theta^2}{12} = \frac{4}{9} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{27} \Rightarrow \sigma = \frac{\theta}{3\sqrt{3}}$$

$$\sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\tilde{\theta}_1 \frac{1}{3\sqrt{3}}} \rightsquigarrow N(0,1)$$

$$P\left(t_1 < \sqrt{n}\,\frac{\tilde{\theta}_1 - \theta}{\tilde{\theta}_1 \cdot \frac{1}{3\sqrt{3}}} < t_2\right) = \beta \qquad t_{1,2} = \mp 1{,}96$$

$$\frac{t_1 \tilde{\theta}_1}{3\sqrt{3}\sqrt{n}} < \tilde{\theta}_1 - \theta < \frac{t_2 \tilde{\theta}_1}{3\sqrt{3}\sqrt{n}}$$

$$-\tilde{\theta}_1 + \frac{t_1 \tilde{\theta}_1}{3\sqrt{3n}} < -\theta < -\tilde{\theta}_1 + \frac{t_2 \tilde{\theta}_1}{3\sqrt{3n}}$$

$$\tilde{\theta}_1 - \frac{t_1 \tilde{\theta}_1}{3\sqrt{3n}} > \theta > \tilde{\theta}_1 - \frac{t_2 \tilde{\theta}_1}{3\sqrt{3n}}$$

$$P\left(\tilde{\theta}_1 - \frac{t_2 \tilde{\theta}_1}{3\sqrt{3n}} < \theta < \tilde{\theta}_1 - \frac{t_1 \tilde{\theta}_1}{3\sqrt{3n}}\right) = \beta$$